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LATE CENOZOIC PALEOMAGNETISM AND CHRONOLOGY OF ANDEAN BASINS OF BOLIVIA: EVIDENCE FOR POSSIBLE OROCLINAL BENDING[1]

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ABSTRACT

New paleomagnetic and radioisotopic data are reported from two late Tertiary high-elevation, non-marine basins in the eastern Cordillera of Bolivia. (1) Quebrada Honda, located at 22° S lat., consists of a 300 m thick section containing abundant Santacrucian or Friasian (middle Miocene) fossil mammals. This locality is constrained by mean $^{40}K/^{40}Ar$ ages of 12.83 ± 0.11 Ma and 11.96 ± 0.11 Ma, and the local magnetostratigraphy is correlated to chrons C5AA through C5A on the Magnetic Polarity Time Scale (MPTS). The fossil mammals from Quebrada Honda have an extrapolated age of about 13.0 to 12.7 Ma. (2) Micaña, located at 17° S lat., consists of a 205 m thick section containing late Miocene fossil mammals, including a megatheriid sloth and the tiny mesothere *Microtypotherium* cf. *M. choquecotense*. This locality is constrained by a fission-track age determination of 6.9 ± 1.1 Ma, and the local magnetostratigraphy is correlated to Chron 7 on the MPTS. The fossil mammals from this section have an extrapolated age of about 7.3 to 7.4 Ma. In conjunction with two other published data sets (Ocros, 13° S, and Salla, 17° S), these late Tertiary Andean localities indicate counterclockwise rotation at 13° S, negligible rotation at 17° S, and clockwise rotation at 22° S. These data could represent local small-block rotations on a scale greater than about 6 km^2. However, these data are also consistent with a model of late Neogene bending of the Bolivian Orocline.

INTRODUCTION

The extensive series of high-elevation Andean basins of Bolivia provide an excellent opportunity for an enhanced understanding of Cenozoic geological history in South America. These non-marine basins are frequently highly fossiliferous, and therefore preserve an excellent record of the unique vertebrate fauna that evolved on that continent during the Cenozoic. In addition, these sedimentary basins oftentimes consist of relatively thick and generally continuous sequences, with numerous interbedded volcanic ash horizons, all of which are necessary prerequisites for combined paleomagnetic and radioisotopic dating studies. Situated in a region of major tectonic activity, these basins are relevant to an understanding of the development of the Andean orogen.

The purpose of this paper is to present new paleomagnetic and geochronological data for two late Cenozoic fossiliferous basins in Bolivia, i.e., Quebrada Honda and Micaña (fig. 1). In addition, interesting new fossil mammals discovered from Micaña will be discussed briefly. The relevant paleomagnetic data from these localities will then be combined with previous results from Salla, Bolivia (MacFadden et al. 1985) to discuss two possible tectonic interpretations: (1) local small-block rotation or (2) late Cenozoic oroclinal bending in the Cordillera Oriental of the Andes.

PALEOMAGNETIC PROCEDURES AND ANALYSES

The field and laboratory regimes were generally the same at Quebrada Honda, Micaña, and Salla and are therefore described here, rather than repeated below in the sections describing each of these localities.

As is standard in magnetostratigraphic studies, three separately oriented hand samples were collected from each field site (horizon). The average spacing between sites was about 5 to 6 m and, because of relatively uniform, fine-grained lithologies, there were no large stratigraphic gaps in sampling. This

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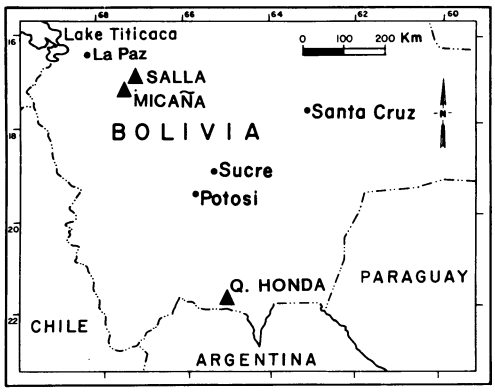


FIG. 1.—Location of the three sites discussed in this report: Salla, Micaña, and Quebrada Honda.

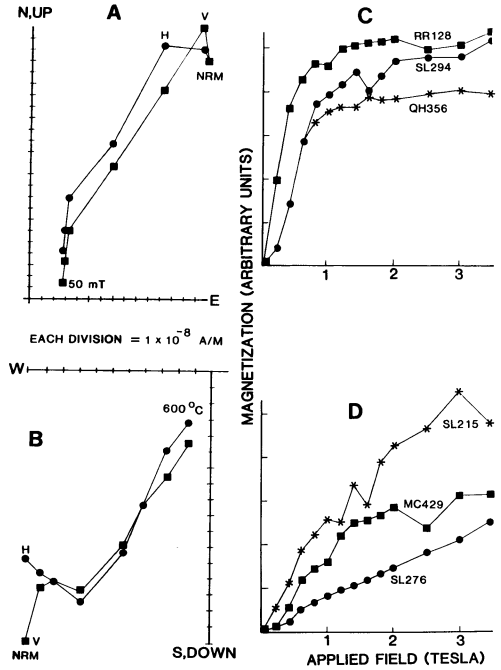


FIG. 2.—Paleomagnetic characteristics for selected samples from the sites listed in figure 1. Orthogonal vector diagrams showing examples of stable, linear decay upon alternating field (*A*) and thermal demagnetization (*B*). H and V represent, respectively, the horizontal and vertical vector components. Isothermal remanence experiments indicate a potentially complex magnetic mineralogy in which some samples saturate in low applied fields of about 1 T (Tesla) or less (*C*), whereas others do not saturate in higher applied fields up to 3 T (*D*). Sample abbreviations are as follows: MC, Micaña; QH, Quebrada Honda; RR, Rio Rosario; SL, Salla.

sampling regime throughout continuous sedimentary sections spanning at least 1 m.y. or more has the advantage of a time averaged view of the ancient geomagnetic field rather than an instantaneous (virtual) geomagnetic pole as is sometimes reported for individual volcanic units.

In the laboratory, the paleomagnetic samples were then cut into 2.5 cm cubes and analyzed in a cryogenic magnetometer (Fuller et al. 1985) at the University of Florida. Paleomagnetic specimens analyzed after 1985 were kept in a newly built, magnetically shielded room (Scott and Froelich 1985) during measurement and demagnetization to attenuate the effects of the ambient magnetic field.

IRM (isothermal remanent magnetization) experiments indicated that a potentially complex coercivity spectrum could be encountered from any site (fig. 2*C,D*). Accordingly, for each site one sample was AF (alternating field) demagnetized and another was thermally demagnetized; the third sample was then demagnetized using either AF or thermal techniques depending upon the results of the first two samples. In most cases samples were demagnetized in a series of between four to ten steps. In some specimens, additional steps were done to isolate characteristic directions of magnetization.

All paleomagnetic laboratory results were subjected to the following standard statistical filtering. Linear trajectories, which are interpreted to represent stable characteristic directions, were chosen from segments of orthogonal vector diagrams (fig. 2*A,B*) produced from the stepwise demagnetization described above. These segments were then analyzed using principal component analysis (Kirschvink 1980) to obtain a least-squares, best-fitting magnetic vector for each sample. Principal component directions were then combined within sites and analyzed using standard Fisher statistics (Fisher 1953) to determine site mean directions and associated statistical parameters. Principal component directions with MAD (maximum angular deviation) values exceeding 20 appeared to yield unreliable results and were not used in further analyses. In some sites where principal component analyses yielded less than three separate, acceptable directions, a single step (usually between 45-50 mT or 400–550°

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C) for a third sample was added to the principal component directions so that $N = 3$, which is the minimum number allowable for Fisher statistics. Site mean directions resulting from this analysis that had the R statistic below critical values (Watson, in Irving 1964, table 4.4) were rejected for further consideration. Only statistically significant (Class I *sensu* Opdyke et al. 1977) site directions were then combined for overall formational mean directions. Both Class I and Class III sites (usually N-1 samples with concordant directions) were used to interpret magnetostratigraphy.

In the stratigraphic sections studied here the sediments are either flat-lying (Quebrada Honda, Salla) or have very shallow dips (usually $\leq 7°$ at Micaña). It was not possible, therefore, to do a fold-test to determine stability and time of acquisition of magnetization. However, the results from these sites indicate overlapping normal and reversed populations, suggesting an ancient, stable magnetization (fig. 3).

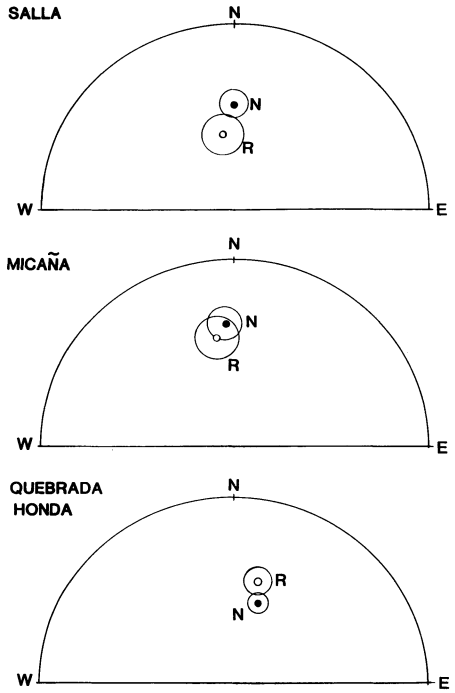


FIG. 3.—Stereographic projections of the statistically significant (i.e., Class I), overlapping normal (N) and reversed (R) populations of site means from the three basins suggest the presence of stable, ancient magnetizations. The circles surrounding the N and R populations represent the 95% cones of confidence.

QUEBRADA HONDA

Quebrada Honda is located on the physiographic boundary between the eastern Cordillera and Altiplano in southernmost Bolivia at lat. 21° 57′ S, long. 65° 25′W at an elevation of about 3500 m (Tojo Sheet, Hoja 6528, 1:50,000 topographic series). This locality was originally described (Hoffstetter 1977) because of its interesting middle Miocene fauna consisting of five orders and 11 families of mammals. Subsequent papers on this locality have discussed stratigraphy and additions to the mammalian fauna (MacFadden and Wolff 1981), as well as descriptive papers on Astrapotheria (Frailey 1987), and Edentata (Frailey 1988). The age of this fauna is either Santacrucian, or Friasian (see below).

During our field work we also discovered another fossiliferous section, Rio Rosario, located some 6 km N of Quebrada Honda. The Rio Rosario section is included here because it contains important fossil mammal horizons and yielded valid paleomagnetic results. McFadden (1989) presented a preliminary discussion of the paleomagnetism of Quebrada Honda and Rio Rosario in a regional tectonic context.

Preliminary Biostratigraphy.—Faunal lists of mammals from Quebrada Honda have been presented elsewhere in the literature (Hoffstetter 1977; MacFadden and Wolff 1981), and those from Rio Rosario are currently under study, principally by R. G. Wolff of the University of Florida. Our field work indicates that fossil mammals are not uniformly dispersed throughout the sections. At Quebrada Honda, fossils occur mostly in a zone between about 25 to 82 m within our composite measured section (fig. 4). At Rio Rosario, fossil mammals occur mostly in the lower 100 m of our measured section (fig. 5). It is unclear, however, whether these fossil localities in the two separate sections represent the same occurrence, i.e., they may or may not be contemporaneous; they do seem to represent the same land-mammal age.

Hoffstetter (1977) and MacFadden and Wolff (1981) assign a Friasian age to the fossil mammals from Quebrada Honda, whereas Frailey (1988) suggests Santacrucian affinities based on the biochronology of the edentates.

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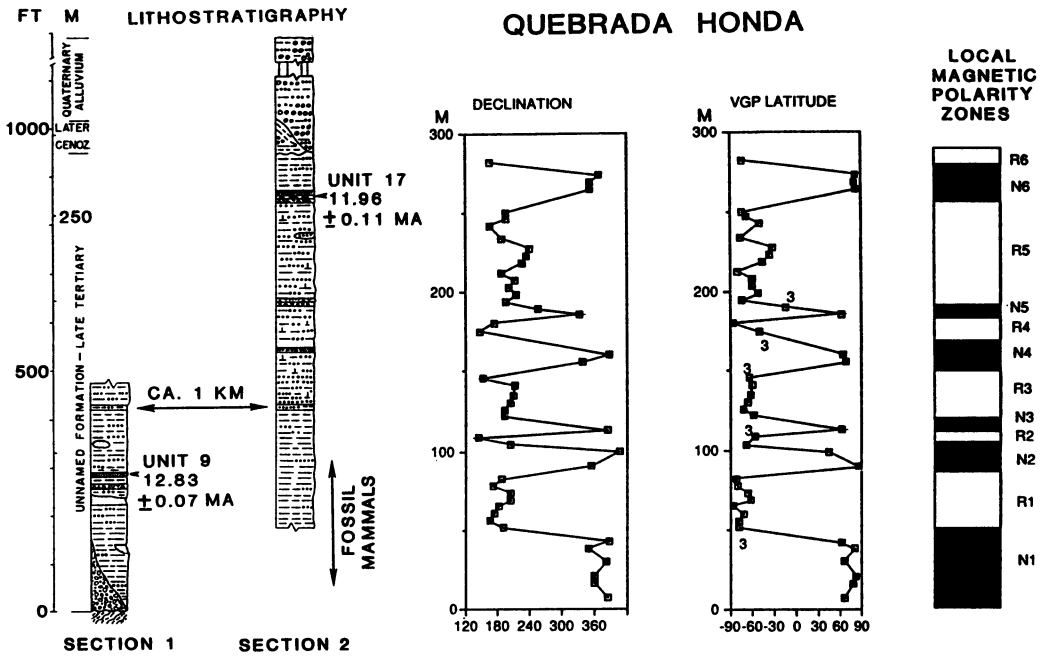


FIG. 4.—Lithostratigraphy, location of fossil mammal horizons, $^{40}K/^{40}Ar$ age determinations, and paleomagnetic results for the Quebrada Honda section. All paleomagnetic sites are Class I, except those with a "3" on VGP plots, which represent Class III (*sensu* Opdyke et al. 1977).

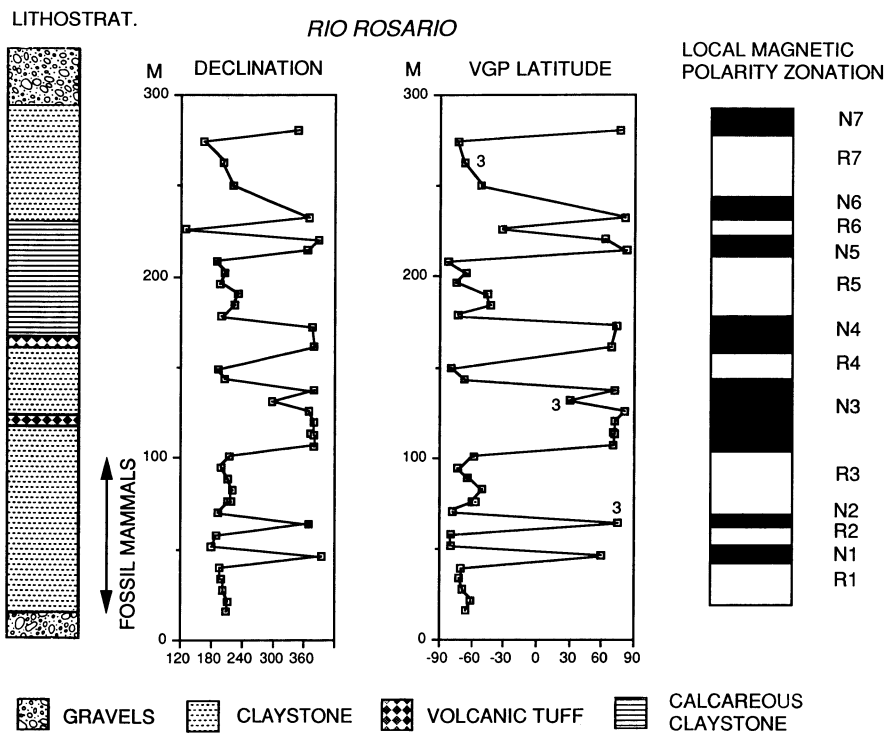


FIG. 5.—Lithostratigraphy, location of fossil mammal horizons, and paleomagnetic results for the Rio Rosario section. All paleomagnetic sites are Class I, except those with a "3" on VGP plots, which represent Class III (*sensu* Opdyke et al. 1977).

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TABLE 1

FISHER STATISTICS FOR SUBSETS OF THE CLASS I DATA FROM QUEBRADA HONDA AND RIO ROSARIO

Subset	Declination (°)	Inclination (°)	R	K	α_{95} (°)	N	ΔD (°)
Quebrada Honda section 1	12.0	−38.1	24.45	16.1	7.3	26	7.4
Quebrada Honda section 2	19.3	−45.9	14.71	11.6	11.3	16	13.1
Quebrada Honda sections 1 & 2	14.5	−41.1	39.04	13.8	6.1	42	6.5
Rio Rosario	19.0	−38.7	35.80	29.5	4.4	37	4.5

NOTE.—Reversed sites are inverted to northern hemisphere; R, resultant; K, precision parameter; α_{95}, 95% cone of confidence; N, number of Class I site means; $\Delta D = 0.8$ [arcsin (sin α_{95}/cos I)], from Demarest (1983).

This uncertainty about land-mammal age assignment is not surprising. Current work suggests that the Friasian land mammal age elsewhere in South America is very poorly characterized. Based on new radioisotopic age determinations, Flynn et al. (1989) suggest that faunas assigned to these two ages may overlap in time.

Lithostratigraphy.—Both the Quebrada Honda and Rio Rosario sections have been mapped as belonging to the Honda Group by GEOBOL (Servicio Geológico de Bolivia). These sections consist of about 300 m of principally fine-grained fluviatile sediments with predominant lithologies reddish brown and yellowish-green silty clays, silts, and fine sands (figs. 4, 5). There are also coarser, lenticular channel sands and gravels, but these are localized. Several interbedded tuffaceous horizons occur within both the Quebrada Honda and Rio Rosario sections. Unfortunately, these distinctive and continuous units cannot be traced laterally within the intervening 6 km between Quebrada Honda and Rio Rosario because of lithofacies variations and Quaternary cover. The base of each section consists of about 50 to 100 m of breccia gravels unconformably lying upon Cambro-Ordovician sediments of the Sama and Iscayachi formations. Unconformably overlying our measured sections, there is a shallow cover (tens of meters or less) of Quaternary fine- to coarse-grained alluvium and fluvioglacial deposits.

Paleomagnetism.—Fifty-nine paleomagnetic sites were collected from our measured sections at Quebrada Honda. In the laboratory, these were subjected to the demagnetization and statistical filtering regimes described above. As a result of this analysis, 42 Class I sites yielded an overall mean paleomagnetic direction of dec. 14.5°, inc. −41.1° (reversed sites inverted, table 1). In addition, five Class III sites were added to this data set to interpret a local magnetostratigraphy, thereby resulting in usable data from 80% of the originally sampled sites. These data yielded a local magnetostratigraphy consisting of 12 zones, N1–R6 (fig. 4).

Forty-seven paleomagnetic sites were collected from the measured section at Rio Rosario. After analysis in the laboratory, 37 Class I sites yielded an overall mean paleomagnetic direction of dec. 19.0°, inc. −38.7° (reversed sites inverted, table 1). In addition, four Class III sites were added to interpret a local magnetostratigraphy, thereby resulting in usable data from 87% of the originally sampled sites. These data resulted in a local magnetostratigraphy consisting of 14 magnetozones, R1–N7 (fig. 5).

The local correlation between the Quebrada Honda and Rio Rosario sections, which are separated by about 6 km, is difficult because of several factors including: (1) the lack of demonstrable lateral continuity of persistent units, like the tuffaceous horizons; (2) significant facies changes between sections; (3) radiometric calibrations only available for Quebrada Honda; and (4) the fact that the magnetic polarity time scale is very "busy" during the Miocene, thereby resulting in numerous small polarity events. Nevertheless, due to the flat-lying nature of the sediments and the presence of generally similar fossil mammals in both sections, it is suggested that Quebrada Honda and Rio Rosario are roughly contemporaneous.

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TABLE 2

$^{40}K/^{40}Ar$ Analytical Data for Tuffaceous Samples from Quebrada Honda, Southern Bolivia

Sample	K1 (%)	K2 (%)	$^{40}Ar^a$ (%)	$^{40}Ar^a \times 10^{-10}$ (mole/gram)	Age ± 1σ[b] (Ma)
Unit 9 (82 m)					
biotite	7.555	7.556	72.3	2.0209	15.36 ± .14
biotite	7.555	7.556	73.3	2.0262	15.40 ± .14
					Mean = 15.38 ± .10
sanadine	10.665	10.649	98.4	2.3630	12.74 ± .08
sanadine	10.665	10.649	95.8	2.3951	12.91 ± .12
					Mean = 12.83 ± .07
Unit 17 (255 m)					
biotite	7.163	7.163	84.8	1.4917	11.97 ± .11
biotite	7.163	7.163	64.2	1.4901	11.95 ± .19
					Mean = 11.96 ± .11

[a] Source of errors: $^{40}Ar = 0.50\%$; $^{40}K = 0.50\%$; spike = 0.73%.
[b] $\lambda_B = 4.6930 \times 10^{-9}$; $\lambda_E = 5.8110 \times 10^{-10}$; $^{40}K/K$ (mol) $= 1.167 \times 10^{-4}$.

Geochronology.—Samples for radioisotopic dating were collected from two tuffaceous horizons interbedded within our measured sections at Quebrada Honda. The lower sample, QH-9, occurs 82 m above the base of our measured section 1 (fig. 4). The upper sample, QH-17, comes from the 255 m level within our measured section 2 (fig. 4).

These samples were analyzed using the $^{40}K/^{40}Ar$ method at the geochronology laboratory at the Australian National University. After crushing and then washing to remove fine-grained material, these ash samples were processed using standard mineral separation techniques, i.e., a combination of heavy liquid and magnetic separation, and seiving. Final separates were thoroughly sonified in acetone to remove any traces of organic solvents.

Sample QH-9 sanidine contained several percent cloudy grains. The sample was treated in an ultrasonic bath for 3 min, followed by additional magnetic separation. This entire procedure yielded an essentially pure separate (>99.9%) of clear grains. There remained, however, a few grains containing inclusions. Both biotite samples QH-9 and QH-17 appeared very fresh and consisted of >99.9% pure biotite.

The analytical data for these samples are presented in table 2. Samples were split into separate aliquants for K and Ar determinations. K concentrations were determined by flame photometry. Ar concentrations were determined by isotope dilution using an ^{38}Ar spike having a concentration known to better than 0.75%. The sanidine was heated to greater than 1600°C for 30 min to insure quantitative extraction of Ar.

The interpretation of conventional $^{40}K/^{40}Ar$ results obtained from young air-fall volcanics, although straightforward in terms of Ar systematics, must take into account several possible sources of geological error. Biotite ages can be incorrect as a result of weathering, and both sanidine and biotite ages can be anomalously old due to detrital contamination. The biotite and sanidine analyses for the lower sample, QH-9, yielded mean age determinations of, respectively, 15.38 ± 0.10 Ma and 12.83 ± 0.07 Ma (table 2). As it is unlikely for fresh sanidine to lose Ar under near-surface conditions, we interpret the biotite age to reflect detrital contamination. We therefore choose to accept the younger date on sanidine of 12.83 ± 0.07 Ma as the best estimate for the age of the ash sample QH-9.

The upper sample, QH-17, within our measured section 2, yielded an age of 11.96 ± 0.11 Ma on the biotite separate (table 2). We have no independent radioisotopic check on the age determined for this sample. However, QH-17 lies stratigraphically above QH-9 (fig. 4); therefore the younger age for QH-17 relative to QH-9 is internally consistent with the superpositional relationships of the tuffaceo-

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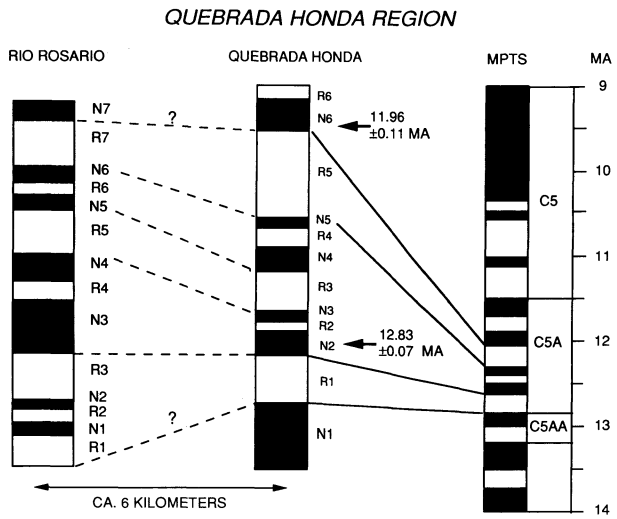


FIG. 6.—Correlation of the Rio Rosario and Quebrada Honda sections to one another and to the Magnetic Polarity Time Scale of Berggren et al. (1985).

ous units from which these samples were collected.

Correlation to the MPTS.—The Quebrada Honda magnetic polarity zonation (figs. 4, 6) is used for correlation to the MPTS because it contains the radioisotopic calibration points. This composite section is dominantly reversed with several normal polarity zones. Given the $^{40}K/^{40}Ar$ ages of 12.83 ± 0.07 Ma at the top of R1 and 11.96 ± 0.11 Ma at the bottom of N6, zone N1 is correlated to the top of Chron C5AA, and R1 to R6 to the lower and middle of C5A (fig. 6). Therefore, using the polarity boundaries of the MPTS, the correlated limits of the Quebrada Honda section are between 13.0 and 11.9 Ma. The fossil localities in zones N1 and R1 thus have an extrapolated age of between 13.0 to 12.7 Ma.

As mentioned above, the correlation of the Rio Rosario section is less secure. Although based on general field relations (i.e., flat-lying and similar elevation) and similar faunas, it is probably contemporaneous with Quebrada Honda. The Rio Rosario section contains 14 magnetozones; therefore there is not a one-to-one match with the 12 zones at Quebrada Honda. Like Quebrada Honda, however, Rio Rosario is dominantly of reversed polarity, and it is correlated to the lower and middle parts of Chron C5A of the MPTS. The fossil mammal localities in the middle of the Rio Rosario section are therefore probably between about 13 to 12.5 Ma.

These correlations of the local sections to the MPTS are not unique, but the one presented here is consistent with the error limits of the radioisotopic calibration points. Given these correlations, zones N3–N6 at Rio Rosario and N2 and N5 at Quebrada Honda have additional reversed intervals (R4 and R6 and R2 and R4, respectively) not recognized on the MPTS.

Overall Paleomagnetic Direction.—The paleomagnetic data for all Class I sites from both Quebrada Honda and Rio Rosario (combined) yielded a mean paleomagnetic direction of dec. 17.8°, inc. −40.7° (reversed sites inverted, table 3). As will be discussed below, these data are of relevance to an understanding of the late Cenozoic tectonic development of the central Andes.

MICAÑA

Micaña is located in the Cordillera Oriental in central Bolivia (fig. 1) at lat. 17° 30′ S, long. 67° W 24′ W at an elevation of about 3800 to 4000 m (Puchuni Sheet, Hoja 6142 III, 1:50,000). Hoffstetter (1986) reported the presence of Pleistocene mammals from this general area (Puchuni), although the actual taxa were not listed. The fossil mammals discovered during our field work and reported below indicate a late Miocene age for the

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TABLE 3

PALEOMAGNETIC DATA FOR THE LOCALITIES FROM BOLIVIA

Locality	Latitude (°S)	Dec (°)	Inc (°)	Ns/Nu	R	α_{95} (°)	K	R (°)	ΔR (°)
Salla	17	353.4	−37.4	104/58	53.7	5.4	13.1	−6.6	5.4
Micaña	17	355.2	−25.8	39/25	24.0	5.9	24.8	−4.8	5.3
Quebrada Honda-Rio Rosario	22	17.8	−40.7	106/79	74.6	3.9	17.9	17.8	4.1

NOTE.—Abbreviations in the heading are as follows: Dec, declination; Inc, inclination; Ns/Nu, number of sites originally sampled/number of sites used for overall mean directions and associated statistics; R, resultant; α_{95}, 95% cone of confidence; K, precision parameter (for R, α_{95}, and K, see Fisher 1953); R, amount of net rotation (negative sign indicates counterclockwise rotation); ΔR, estimated error in calculation of rotation (Beck 1980; Demarest 1983; Beck et al. 1986*a*).

Micaña locality. MacFadden (1989) presented a preliminary discussion of the paleomagnetism of Micaña in a regional tectonic context.

Lithostratigraphy.—The sedimentary section at Micaña consists of an unnamed unit, informally referred to here as the Micaña beds (it was previously mapped as equivalent to the Salla Beds by GEOBOL in their regional mapping program). Our measured section consists of 205 m of strata that generally dip gently (usually about 7°), although a small portion at the base dips as much as 15°. These strata consist of fluviatile and volcaniclastic sediments with several interbedded tuffaceous horizons. The Micaña beds unconformably overlie deformed Paleozoic sediments. The top of our section is bounded by a fault, above which unconformably lie younger sediments, including a dated tuffaceous horizon that produced a minimum age constraining our section (fig. 7).

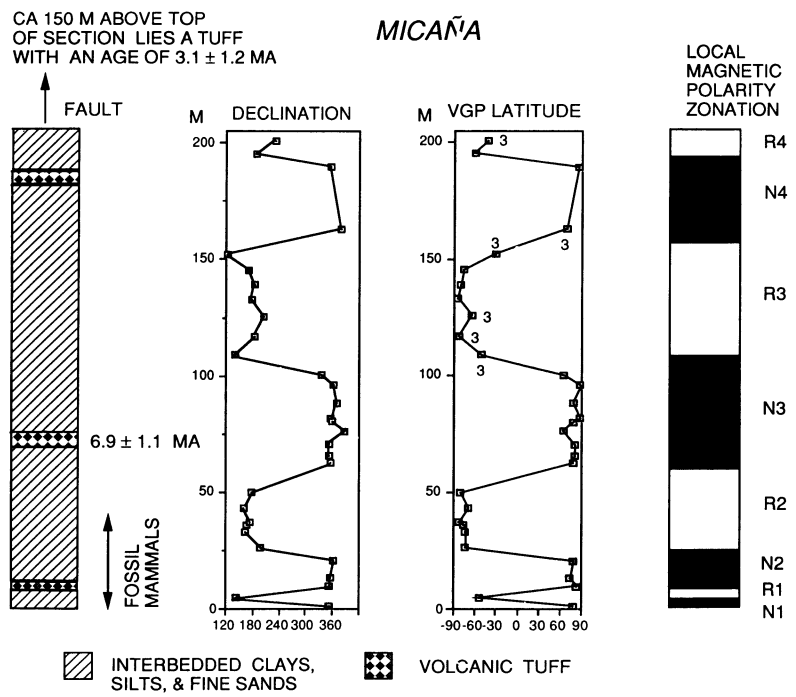


FIG. 7.—Lithostratigraphy, location of fossil mammals, fission-track (zircon) age determinations, and paleomagnetic results for the Micaña section. All paleomagnetic sites are Class I, except those with a "3' on VGP plots, which represent Class III (*sensu* Opdyke et al. 1977).

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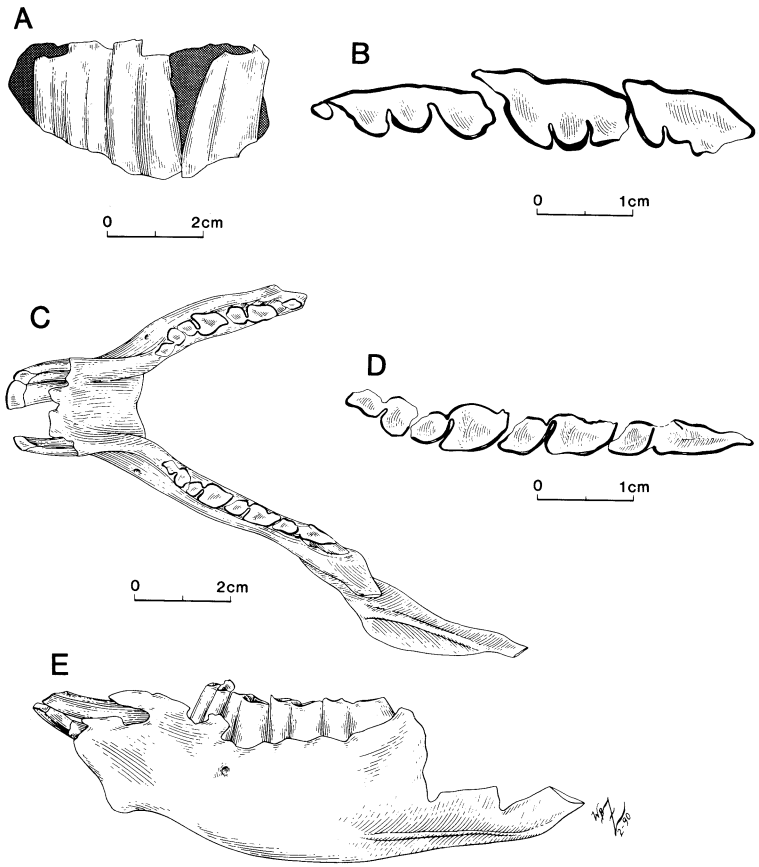


FIG. 8.—*Microtypotherium* cf. *M. choquecotense*, UF 123620, from the late Miocene of Micaña, Bolivia. Right M^1-M^3, lateral (*A*) and occlusal (*B*) views (reversed). Lower mandible, dorsal (*C*), left occlusal (*D*) and left lateral (*E*) views, including P_4-M_3.

Paleontology.—Several fossil mammal specimens were collected in 1986 from the lower 40 m of our measured section. Of these, two taxa are identifiable: the first is a medium-sized megatheriid sloth of indeterminant genus or species, UF (University of Florida Vertebrate Paleontology collection) 123621. The second specimen consists of a relatively well-preserved individual, including a partial upper cheek tooth dentition and the lower jaw (with symphysial and cheek tooth dentitions) of a small notoungulate, UF 123620 (fig. 8). Comparisons made with specimens in the paleontological museum at GEOBOL in LaPaz indicate the UF specimen is most similar to GB 223, referred to *Microtypotherium* cf. *M. choquecotense* Villarroel (1978; also see Villarroel 1974). This small and primitive mesotheriid is otherwise known from the late Miocene of the Bolivian Altiplano, and therefore suggests a similar age for the base of the Micana section.

Paleomagnetism.—Thirty-nine sites were collected from our measured section at Micaña. As for Quebrada Honda, IRM experiments resulted in either low field saturation or no saturation in fields up to 3.5 T, and as described above, both AF and thermal demagnetization were done on samples from each site.

After the laboratory regime and statistical filtering of resulting data (as described above), 25 of the 39 originally sampled sites were statistically significant (Class I, *sensu* Opdyke et al. 1977) and these were used for calculation of an overall paleomagnetic mean direction of dec. 355.2, inc. −25.8° (table 3). Because of relatively shallow dips (mostly about 7°) for the measured section, it was not possible to do a fold test for stability. How-

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TABLE 4

FISSION-TRACK (ZIRCON) AGE DATA FOR SAMPLES COLLECTED FROM MICAÑA, BOLIVIA

Sample	Location	Spontaneous Tracks (ρs) $\times 10^6$ t/cm^2	Induced Tracks (ρi) $\times 10^6$ t/cm^2	Dose $\times 10^{15}$ n/cm^2	T (Ma)	$\pm 2\sigma$ (Ma)	No Grains
Micaña DF-5483	72 m	1.23 (193)	9.39 (739)[a]	.88	6.9	1.1	6
Micaña DF-5484	>205 m	2.74 (33)	4.64 (279)[a]	.88	3.1	1.2	6

[a] ( ) = Number of tracks counted; $\lambda_f = 7.03 \times 10^{-17}$ yr^{-1}.

ever, the overlap of the mean directions for the two populations of normal and reversed polarity (fig. 3) suggest an ancient, stable magnetization.

Although not used for the tectonic interpretations below, six Class III sites were used in addition to the 25 Class I sites for interpretation of the magnetic polarity stratigraphy. Therefore 31 of the 39, or 80%, of the originally sampled sites produced usable results for establishment of a magnetic polarity zonation. The magnetic data from the lower 160 m of our measured section are generally excellent and resulted in the recognition of six polarity zones designated here as N1–R3. However, two of these (N1 and R1), located at the base of the measured section, are each defined by only a single site, and should therefore be considered with caution when using the local polarity pattern for correlation to the MPTS. The magnetic data represented by zones N4 and R4 from the top of the measured section are less secure, and in this zone three superposed sites between 170 through 182 m were rejected because of poor data.

Geochronology.—Samples for fission-track age determination were collected from tuffaceous horizons within our measured section. Mineral separations were done at Dartmouth and then sent to the U.S. Geological Survey in Denver for further analysis. The analytical data from these samples (table 4) was determined by the external detector method (Naeser 1979).

Zircon grains were mounted in FEP teflon at 300°C, then were polished, and spontaneous tracks were counted. The zircons were then etched for 34 hr in a eutectic melt of KOH and NaOH at 215°C to develop the tracks (Gleadow et al. 1976), and were then covered with an external muscovite detector to record induced fission tracks. These samples were irradiated with NBS glass standard 962 in the USGS reactor at Denver. Following irradiation, the muscovite detector was etched in 48% HF at 20°C for 15 min. The neutron dose was determined by counting the number of tracks in the muscovite detector, which covered the glass standard during irradiation. The NBS Cu calibration was used to determine the dose. The error reported in table 4 was determined by Poisson statistics based on the total number of tracks counted.

Correlation to the MPTS.—The minimum age of the Micaña beds is constrained by a fission-track age of 3.1 ± 1.2 Ma, which lies about 150 m stratigraphically above the faulted top of our measured section (fig. 9). The lower part of the Micaña beds is con-

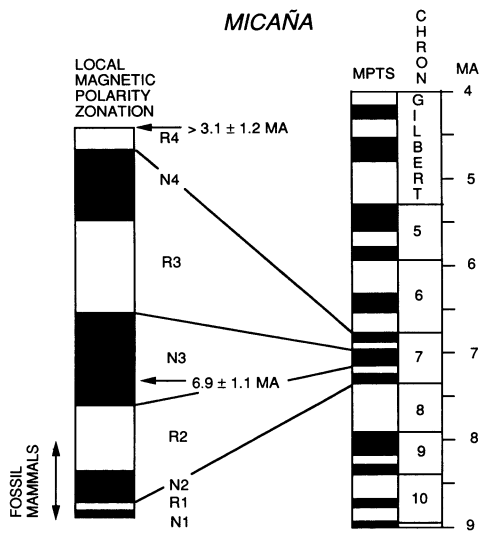


FIG. 9.—Correlation of the Micaña magnetic polarity zonation to the Magnetic Polarity Time Scale of Berggren et al. (1985).

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strained by a fission-track age determination of 6.9 ± 1.1 Ma, which lies at 72 m in our measured section (table 4). Using these data, it is difficult to correlate unambiguously the local polarity pattern to the MPTS. If the mean age of 6.9 Ma is considered, however, then the normal zone in which it lies (N3) can be correlated to the middle of Chron 7 of the MPTS. Following this match, the underlying zones N2 and R2 and overlying zones R3 to R4 probably correlate to, respectively, the bottom and top of Chron 7. As mentioned above, zones N1 and R1 are both represented by only one site each, and we do not consider these to be a reliable basis for correlation to the MPTS. Also, the upper 10 m of Micaña beds represented by R4 may represent the bottom of Chron 6 although, as mentioned above, the magnetic data are poor from this part of the section. If these correlations are correct (excluding the problematical zones N1 and R1), then the fossil mammals collected from the lower 40 m of section lie within the bottom of Chron 7; these therefore have an age extrapolated from the MPTS to be about 7.4–7.3 Ma and are certainly older than the overlying 6.9 ± 1.1 Ma age determination.

SALLA

Salla is located in the Cordillera Oriental in central Bolivia about 100 km southeast of La Paz (fig. 1) at lat. 17° 11′ S, long. 67° 39′ W at an elevation of about 3485 to 4085 m. Since its discovery in 1963, this locality has been worked extensively by field crews from the University of La Paz (Bolivia), Musée National de Paris (France), GEOBOL, Kyoto University (Japan), and the University of Florida (USA). In addition to extensive Deseadan mammalian fauna recovered during this field work, Salla has attracted much attention in the literature because the oldest South American platyrrhine primate, *Branisella,* occurs there. Numerous papers deal with various aspects of this site and are far too numerous to list here; these are referenced elsewhere (e.g., MacFadden et al. 1985; MacFadden 1990).

Of relevance to the tectonic discussion below, MacFadden et al. (1985) presented the magnetostratigraphy of the Salla Beds, and additional radioisotopic age determinations are presented by Naeser et al. (1987). Based on results from 10 (of 19 originally collected) paleomagnetic sites, Hayashida et al. (1984) suggested about 25° of counterclockwise rotation for the Salla Beds. These results, however, are different from those presented in MacFadden et al. (1985) and below.

MacFadden et al. (1985) collected 104 individual sites from throughout the Salla Beds, and after standard laboratory treatment, 76 of these were used to interpret the local magnetic polarity stratigraphy. For the present study, only statistically significant sites (Class I, *sensu* Opdyke et al. 1977) are used, and those with intermediate directions (near 90 or 270°) and/or shallow inclinations ($< \pm 10°$) were rejected from further tectonic interpretations. As indicated in table 3, the resulting 58 sites (reversed sites inverted) yielded an overall paleomagnetic mean direction of dec. 353.4, inc. −37.4°.

TECTONIC INTERPRETATIONS: LOCAL ROTATION OR OROCLINAL BENDING?

Our paleomagnetic data for Salla, Micaña, and Quebrada Honda (table 3) are relevant to an understanding of the late Cenozoic tectonic history of the Cordillera Oriental of the central Andes. Given the associated confidence intervals, the first two localities are unrotated in any sense, whereas Quebrada Honda has undergone significant clockwise rotation about a vertical axis. Beck (1989) discusses seven potential geological processes that can result in declination anomalies. With regard to the sites presented here for Bolivia, there are two possible interpretations for the observed paleomagnetic directions; these data could represent either (1) local (i.e., small-block) structural deformation or (2) oroclinal rotation.

1. Local Structural Rotation.—Paleomagnetic vector rotations, also termed declination anomalies, have been useful in determining rotations of structural blocks in many tectonically active settings. For example, it is known that within the structurally complex Mojave desert of southern California, individual blocks of some 10 to 100 km^2 have undergone independent, late Cenozoic rotations (e.g., Luyendyk and Hornafius 1987; Ross et al. 1989; MacFadden et al. 1990). Larger blocks on the order of a few 100 km^2 are also suggested to have undergone block rotation (Luyendyk 1989). A clear pattern

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predicting rotation directions has not yet emerged in and adjacent to the Mojave block, but these are probably related to the kinematics of genetically similar faults bounding the various local structural blocks.

Beck (1988) analyzed late Jurassic to Recent paleomagnetic data from the active margin of western South America. He concluded that (also see Beck 1987; Beck et al. 1986*b*), contrary to previous claims supporting oroclinal bending (e.g., paleomagnetic data summarized in Isacks 1988), the observed declination anomalies could support a mechanism of in situ small-block rotation of local crustal blocks caused by regional shear. Beck (1988) also realizes that some of the paleomagnetic data may support the Bolivian Orocline hypothesis. On the other hand, in a study of early Jurassic to early Cretaceous paleomagnetic data from northern Chile, Hartley et al. (1988) attribute all observed declination anomalies to local block rotations within the Andean forearc, although the exact tectonic mechanism causing this pattern is still uncertain.

Of the three paleomagnetic data sets presented in table 3, only the one from Quebrada Honda/Rio Rosario shows significant rotation, with a clockwise declination anomaly of 17.8° ± 4.1°. If this rotation represents local structural rotation, at what scale did this occur? This question can be partially addressed by examining the subsets of paleomagnetic data presented in table 1. Within our Quebrada Honda composite, two paleomagnetic data sets were collected from sections 1 and 2 (fig. 4), which are separated by about 1 km, and the general field area covered is about 1.0 km^2. At a larger scale, the Quebrada Honda composite and Rio Rosario sections are separated by about 6 km and encompass an area of about 6 km^2. As shown in figure 10, the paleomagnetic data sets both from within sections 1 and 2 at Quebrada Honda and between Quebrada Honda and Rio Rosario are indistinguishable, indicating no differential rotation at a scale between 1.0 and 6.0 km^2. If non-oroclinal structural rotation caused the observed declination anomaly for the combined Quebrada Honda/Rio Rosario data sets, then it must have occurred on a scale greater than 6 km^2; this is certainly plausible using some mesoscale tectonic processes described in Beck (1989). In addition, because

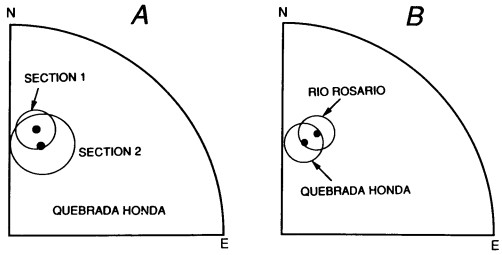


FIG. 10.—Overlap of the mean directions for Class I sites for sections 1 and 2 at Quebrada Honda (*A*) and between Quebrada Honda and Rio Rosario (*B*). Circles represent α_{95}, the 95% cones of confidence.

these sediments are flat-lying, any local structural rotation could only have occurred about a vertical axis; in areas such as the Mojave desert where local block rotation is known to exist, Miocene sediments are rotated about both vertical and horizontal axes. It is less clear how small-block rotation could have resulted in a sedimentary sequence that is neither extensively faulted nor tilted.

If however, the new data presented here do not represent local/regional block rotation, then they may constrain both the timing and mechanism of development of the Bolivian Orocline.

2. Oroclinal Rotation.—Carey (1955) proposed the concept of an orocline to describe regional tectonic bending of mountain ranges. He coined the term "Bolivian Orocline" (Carey 1958) to describe the flexure of roughly 45° seen in the Andes, with the axis of bending aligned SW-NE along the "Arica deflection." Assuming that this is a valid tectonic model, the exact mechanism and age of bending of the Bolivian orocline has remained somewhat enigmatic. Carey (1955) suggested that the flexure of 45° can be explained by bending of the north limb of the orocline above the Arica deflection, whereas some paleomagnetic data suggest rotation of the southern limb (Palmer et al. 1980). On the other hand, Isacks (1988) suggests roughly equal amounts of bending on both limbs of the orocline. Isacks' model is confirmed by relevant paleomagnetic data (summarized in Isacks 1988), most of which are based on Mesozoic rocks and therefore only constrain bending as post-Cretaceous. If the principal phase of oroclinal bending is related to sub-

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duction of the Nazca plate and Andean uplift (Allenby 1987; Isacks 1988), then rotation should have occurred much later, i.e., during the late Cenozoic. However, the lack of paleomagnetic studies of young rocks in relevant areas has prevented testing this possibility.

In a previous study of Neogene rocks in the Andes, Heki et al. (1985) determined $-14.5 \pm 5.5°$ (negative indicates counterclockwise) rotation for the Ocros dyke swarm in southern Peru, which is located to the north of the Arica deflection on the presumed northern limb of the orocline. (These data were subsequently recalculated by Beck [1988] as $-14 \pm 26°$, which if correct imply that rotation may be negligible).

With regard to the new paleomagnetic data from Bolivia, if declinations do not represent local small-block rotation, then the results from Salla, Micaña, and particularly Quebrada Honda (table 3) constrain the timing, mechanism, and areal extent of oroclinal bending. The rotational statistics using the overall paleomagnetic mean directions and associated confidence intervals are taken from Beck (1980), Beck et al. (1986*a*), and Demarest (1983). Heki et al. (1985) determined that the late Tertiary pole is indistinguishable from the modern pole, and therefore all observed declination anomalies are interpreted to represent rotation (about a vertical axis) relative to cratonic South America.

The overall paleomagnetic mean directions for Salla and Micaña are, respectively, $-6.6 \pm 5.4°$ and $-4.8 \pm 5.3°$, which suggest slight counterclockwise or (if the error is considered) negligible rotation for these sites. Isacks' (1988) model predicts little rotation near the axis of the orocline. Located at 17° S, Salla and Micaña are near this axis (if projected inland from the Arica deflection) and are therefore consistent with Isacks' model of roughly equal amounts of bending on both limbs of the orocline (fig. 11). In contrast, the paleomagnetic data from Quebrada Honda at 22° S suggest $17.8 \pm 4.1°$ of clockwise rotation south of the Arica deflection. These results from the presumed southern limb are of similar magnitude (although of opposite direction) to those from the Neogene Ocros dyke swarm in Peru (using the original data set of Heki et al. 1985), located at 13°, and suggest counterclockwise rotation on the northern limb of the Bolivian orocline (figs. 11, 12).



FIG. 11.—Map of central part of South America showing general extent of the Andes (shaded) and the three sites studied here as well as the results from the Neogene Ocros dyke swarms of Peru (Heki et al. 1985). The arrows within the half-circles represent the amount of rotation (R) and the black cones encircling the arrows represent the ΔR, or estimated error (Beck 1980; Demarest 1983; Beck et al. 1986*a*).

The exact age of the Ocros dyke swarm is unknown, other than that it is late Miocene to Pliocene (Heki et al. 1985). On the other hand, the radioisotopic ages and correlation to the MPTS for Quebrada Honda (fig. 6) constrain the rotation in this region to post-12 Ma. There are no discernable trends in declination anomalies (figs. 4, 5) throughout the stratigraphic sections at Quebrada Honda

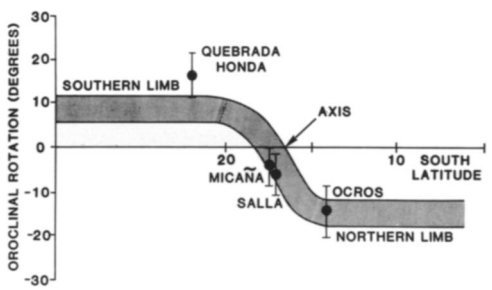


FIG. 12.—Predicted values (shaded) of rotation for a model of oroclinal formation (Isacks 1988) that assumes roughly equal amounts of rotation on the northern and southern limbs of the orocline. The new data from this study and those presented for the Neogene Ocros dyke swarm of Peru (Heki et al. 1985) are consistent with this model and suggest that oroclinal rotation occurred during the late Cenozoic, i.e., post-12 Ma. The bars on either side of the calculated declination rotation (R) represent the estimated error (ΔR, see references cited in fig. 11).

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(and Rio Rosario), so all observed rotation seems to be post-depositional.

If the observed data do not represent local small-block rotation, then the declination anomalies determined from the non-axial portions of the Bolivian orocline are post-depositional and occurred during the late Cenozoic, after about 12 Ma. If the magnitude of the rotations from Orcos and Quebrada Honda are summed, then there has been about 32° of net rotation during this time. This, however, does not account for all of the rotation required to form the 45° bend represented by the Arica deflection. The late Mesozoic data compiled in Isacks (1988) similarly does not account for the additional 15° of rotation. As Beck (1988) has suggested, this would imply an earlier phase of oroclinal development during the middle Mesozoic.

We prefer the interpretation that our data suggest late Neogene oroclinal bending because it seems consistent with the model proposed by Isacks (1988, and the paleomagnetic data compiled therein). However, we cannot reject the possibility that the new data presented here may represent small-block rotations on a scale greater than about 6 km^2. Additional paleomagnetic data from other late Cenozoic rocks in relevant areas of South America are needed to understand better rotational histories related to the Andean orogen and whether these tectonic regimes resulted from small-block rotations or oroclinal bending.

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